Identifying Key Predictors of Cardiovascular Diseases

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Introduction

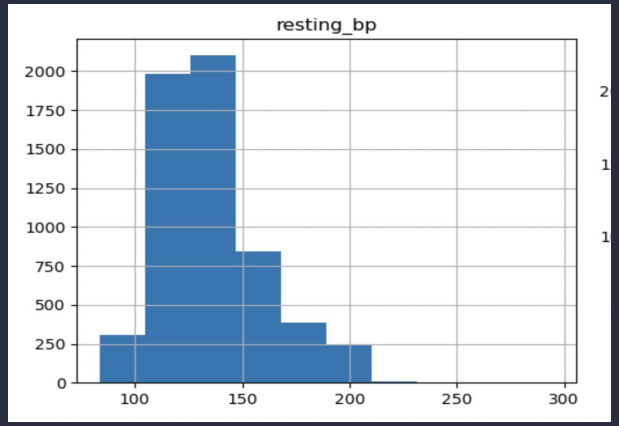
The goal of this project is to analyze the datasets related to cardiovascular health including heart disease, heart failure, and general cardiovascular disease. Although the datasets focus on slightly different conditions, they contain many similar health indicators such as age, blood pressure, cholesterol levels, gender, and heart rate. Using these, I plan on identifying the most significant predictors associated with these conditions.

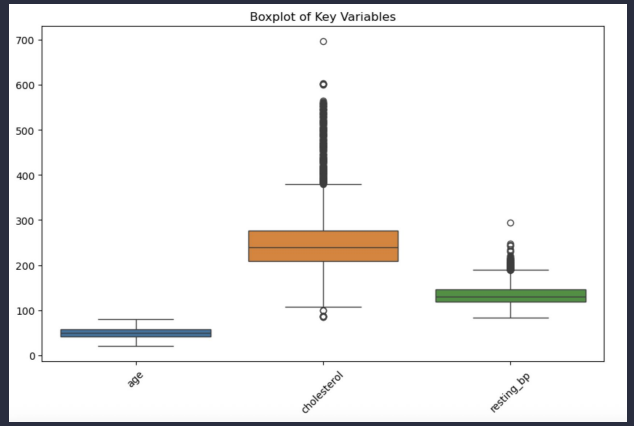
Data Sources

Data Source	Description	List of Fields	Data Size
Framingham Heart Disease Dataset	Study conducted in 1948 through 1952 in FramingHam, Massachusetts to investigate risk factors of CVD	Age, sex, education, currentSmoker, cigsperday, BPmeds, stroke, hyp, diabeties, chol, sysBP, diaBP, BMI, heartrate, glucose, tenyearCHD	4240
Heart Failure Prediction Dataset	Combined datasets sourced from Cleveland, Hungary, Switzerland, Long Beach and Stagalog Dataset	Age, sex, chest pain type, resting BP, cholesterol, fasting blood sugar, resting ECG, max heart rate, exercise angina, oldpeak, ST slope, target	918
CardioVascular Disease Dataset	Values collected at the moment of medical examination with three types of input information	Age, patientID, chestpain, bp, restingBP, chol, sugar, electro, maxheartrate, slope, vessels, exercise, oldpeak, vessels target	1000

Visualizations

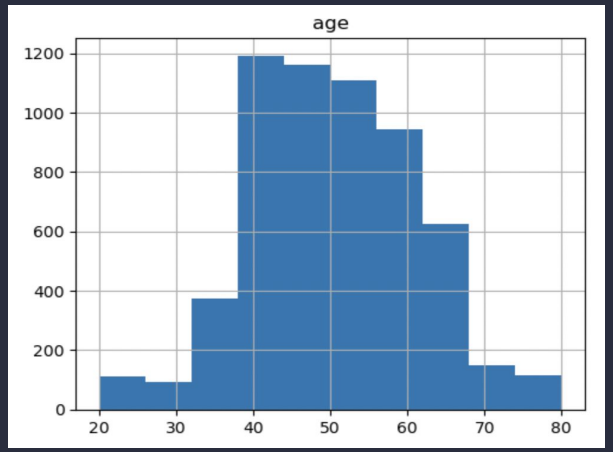
Resting BP
Mean: 135
Min: 84
Max: 295
Std: 24

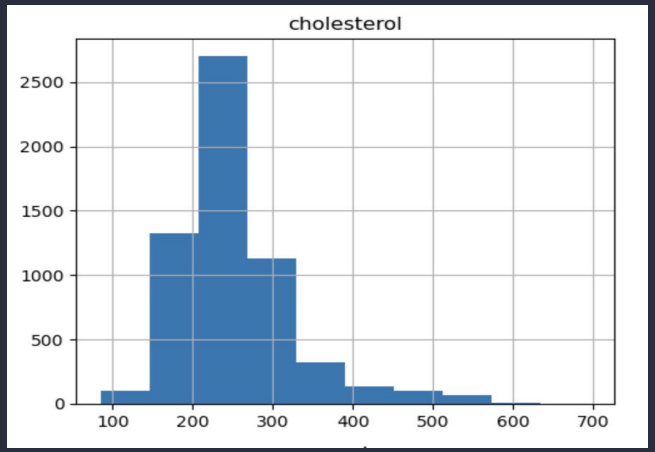




Sex
Male: 3100
Female: 2782
Total Count: 5882

Age
Mean: 50
Min: 20
Max: 80
Std: 10.78





Cholesterol
Mean: 253
Min: 85
Max: 696
Std: 71

Model Assumptions

Independence

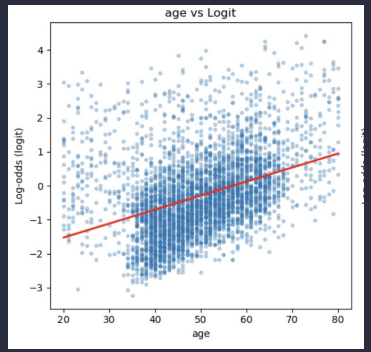
- Since each value is a unique patient with no repeated measurements ,the study is independent and no test is required

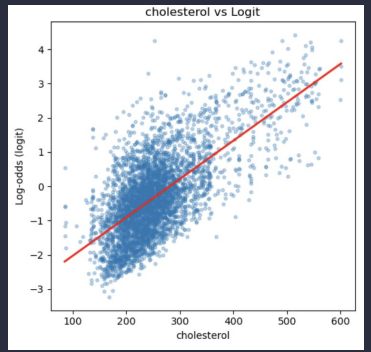
Multicollinearity

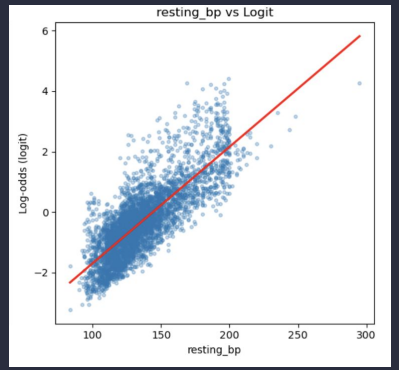
age	sex	cholesterol	resting BP
1.053269	1.004377	1.115329	1.147089

All <5, variables are independent of each other

Linearity







No curve - linearity met

Final Model

Logistic Model
- Binary outcome (Risk of diseases yes/no)

Predictors
- sex, age, cholesterol, resting blood pressure

Likelihood Ratio test
- p-value < 0.001 meaning at least one predictor is significant

Wald test
- All four predictors had a p-value < 0.001

Odds Ratio

Feature	Odds Ratio	95% Interval
Resting BP	2.10	[1.93, 2.29]
cholesterol	1.81	[1.67, 1.97]
sex	1.62	[1.49, 1.77]
age	1.23	[1.14, 1.33]

Resting blood pressure is the highest predictor of heart disease!

Model Performance

Metric	Value
Accuracy	0.516
Recall	0.849
Precision	0.300
F1 score	0.433
AUC	0.764

- The model is pretty strong with an AUC score of 0.764
- Since this is medical data recall should be prioritised over precision since missing a heart disease case carries a greater risk than a false positive

Challenges

Combining three datasets

- Resulted in only four common features available for analysis, stronger predictors such as chest pain and heart rate could not be used
- Had to eliminate heart rate predictor entirely as one of the datasets recorded patients maximum heart rate leading to inconsistent summary statistics
- Common features had different variable names, needing to standardize for consistency

Model

- Unequal distribution of heart disease cases, needing to reweight and use a threshold of 0.3 to improve recall
- Added interaction terms such as age x blood pressure and sex x age with no improvement in AUC score

Thank you!